

Approach to Regional Anaesthesia

The Big Picture

Regional anaesthesia = block the nerves, skip the general. Ideal for **limbs, lower abdomen, Caesarean**, and patients with **major heart/lung disease** who'd be high-risk under GA. But it's never 100% — always have a **plan B (GA)**. Key dangers to know: **neuraxial complications** and **local-anaesthetic systemic toxicity (LAST)**.

What Counts as Regional

- **Neuraxial**: spinal (abdomen/pelvis/lower limb), epidural (neck→lower limb), caudal (perineal).
- **IV regional block (Bier)** — upper/lower limb.
- **Peripheral nerve blocks** — interscalene (shoulder/elbow), axillary/supra-/infraclavicular (arm), **sciatic** (most of leg), **femoral** (rest of leg).

Why Use It

- Immobility + analgesia without GA risks.
- Great for distal-extremity surgery without sedation — attractive in severe pulmonary/cardiac/renal disease.
- Can complement GA for **lasting post-op pain relief**.
- **Spinal/epidural = technique of choice for Caesarean**.

Safe Practice (the checklist)

Consent + preparation, equipment, **monitoring**, trained assistance, environment, documentation. Requires anatomy knowledge, manual dexterity, resuscitation skills, and **GA capability as backup**.

Neuraxial Contraindications

- **Absolute**: patient refusal, **anticoagulation**, local sepsis, raised ICP, (true amide allergy — rare).
- **Relative**: severe valvular stenosis/Eisenmenger/peripartum cardiomyopathy, hypovolaemia, expected haemorrhage, previous back surgery, neurological disease, systemic sepsis.

Neuraxial Complications

- **Hypotension**, bradycardia (mid-thoracic block → can arrest), **high/total spinal** (compromises breathing), urinary retention, nerve damage.
- **PDPH** (post-dural-puncture headache), infection (abscess, meningitis), **spinal canal haematoma** (worse with coagulopathy → can cause permanent paraplegia).
- Serious (death/paraplegia) ~**1:50,000–1:140,000**.

LAST (Local Anaesthetic Systemic Toxicity)

- **Cause**: too much LA into circulation. **Avoid** by slow injection, small boluses, frequent aspiration.
- **Presentation (progressive)**: light-headedness, **peri-oral tingling**, **metallic taste**, **tinnitus**, confusion, tremors → **convulsions**, **coma** → bradycardia, hypotension, **CVS collapse**, **respiratory arrest**.
- **Management**: stop injection; **ABC + 100% O₂**; intubate/ventilate; treat seizures (midazolam/propofol/thiopental); **IV 20% lipid emulsion (Intralipid) 1.5 mL/kg bolus** then infusion; CPR/ALS if pulseless.

Allergy

Rare. **Esters** more allergenic than **amides**. Ranges from rash to laryngeal oedema/anaphylaxis.

Choice of Agent

- **Lidocaine** — medium-acting amide, dose **2 mg/kg** (cerebral irritation before cardiac depression); topical airway forms (4%, spray).
- **Bupivacaine** — long-acting amide for post-op analgesia, dose **3 mg/kg (7 with adrenaline)**; **prolonged cardiotoxicity** in high doses.

Adding Epinephrine (1:200,000)

- **Advantages:** limits systemic absorption, prolongs duration, lowers toxicity risk, epidural test dose.
- **AVOID in:** unstable angina, dysrhythmias, uncontrolled HTN, uteroplacental insufficiency, **end-arteries (digits, penis)**, IV regional anaesthesia.

Walk-Away Points

- Regional shines in high-risk cardiac/pulmonary patients + **Caesarean (spinal/epidural)**; always have GA backup.
- Neuraxial absolute CIs: **refusal, anticoagulation, local sepsis, raised ICP**.
- **LAST** → stop, ABC/O■, **lipid emulsion**.
- Bupivacaine = cardiotoxic; **no adrenaline in end-arteries**.

Regional Anaesthesia — Revision Table	
Technique	Use
Spinal	Abdomen, pelvis, lower limb
Epidural	Neck → lower limb; post-op analgesia
Caudal	Perineal
IV regional (Bier)	Upper/lower limb
Interscalene block	Shoulder, elbow
Sciatic / femoral	Leg
Spinal/epidural	Caesarean (choice)
Neuraxial CI	Type
Patient refusal	Absolute
Anticoagulation	Absolute
Local sepsis	Absolute
Raised ICP	Absolute
Severe valve stenosis, hypovolaemia	Relative
Previous back surgery, neuro disease	Relative
Neuraxial complication	Note
Hypotension / bradycardia	Can → arrest (mid-thoracic)
High/total spinal	Compromises breathing
PDPH	Headache
Spinal haematoma	Coagulopathy → paraplegia
Serious (death/paraplegia)	1:50,000–1:140,000
LAST	Detail
Cause	Excess LA in circulation
Early signs	Peri-oral tingling, metallic taste, tinnitus
Late	Convulsions, CVS collapse, resp arrest
Treatment	Stop, ABC/O $\blacksquare$, seizures Rx, 20% lipid emulsion
Agent	Class / dose
Lidocaine	Amide, 2 mg/kg , medium-acting
Bupivacaine	Amide, 3 mg/kg (7+adrenaline) , cardiotoxic
Allergy	Esters > amides
Epinephrine (1:200,000)	
Advantages	↓ absorption, prolongs block, test dose
Avoid	Digits/penis, unstable angina, IV regional, dysrhythmia

Bold = high-yield.